When Deep Hedgers Blow Up: Interpretable Volatility-Surface Prototypes as a Regime-Robust Floor

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Abstract

PPO and SAC deep-RL hedgers blow up by 12–32× on fourteen years of real options data when tested across two markets and two crises; our interpretable volatility-surface prototype is *the only learned policy stable across both seeds and markets.* It matches the strongest classical baseline (delta–vega) on tail risk (CVaR_{95} = 2.34±0.10 across five seeds on SPY) while remaining fully auditable: every hedge is a similarity-weighted blend of named market regimes (calm, elevated-skew, stressed). The failure of deep RL is located in the objective—PPO and SAC share the same anchored action space as our method but maximise mean P&L, which provides no mechanical reason to rotate into safety assets when the VIX triples overnight, and a black-box MLP is seed-unstable for the same reason. The method runs as a bounded residual on the classical delta–vega hedge, trained under a tail-weighted CVaR objective. Cross-market robustness is not free: a naive residual *loses* on the second market, a failure we trace to a model-selection-under-regime-shift artefact that the tail-weighted objective repairs, restoring parity with delta–vega on both held-out markets (Stouffer $p = 0.079$, not significant at $\alpha = 0.05$). The known boundary is the COVID-2020 SPY fold, where the anchored residual can add tail risk before a pre-registered volatility-scaled cap intervenes (repairing CVaR_{95} from 60.0 to 18.0, cap parameters fixed on training data only). To our knowledge, this is the first empirical test of prototype hedging on live options data; within this scope interpretability and tail-risk control are not a trade-off.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics.*

Keywords

deep hedging, explainable AI, prototype models, implied volatility surface, Conditional Value-at-Risk, risk management

ACM Reference Format:

Anonymous Author(s). 2026. When Deep Hedgers Blow Up: Interpretable Volatility-Surface Prototypes as a Regime-Robust Floor. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Hedging means holding offsetting positions so a portfolio barely moves in value when the market does; the hard case is hedging through a violent regime change, where off-the-shelf deep learning fails badly. On the SPY options test period spanning the COVID-2020 crash and the 2022 bear market, a default PPO deep-RL hedger trained on 2010–2018 data reached a CVaR_{95} tail loss of 35.7 (the average loss on the worst 5% of days), twelve times worse than a simple delta–vega hedge (2.84). A SAC hedger reached 89.7, thirty times worse. These policies had not failed to learn in-sample. They had learned *too well*, absorbing the slow directional drift of the 2010–2018 bull market and replicating it out-of-sample as large, unintended directional and volatility bets. When the market regime changed, their "edge" became catastrophic exposure. The black-box MLP was no better (CVaR_{95} = 13.2).

This is a systematic failure mode, not a tuning accident. It recurs across two real option markets (SPY, QQQ), five random seeds, and every year that includes a genuine stress event (Figure 4), yet it is absent from the interpretable prototype hedger we introduce here (CVaR_{95} = 2.34 on SPY). This motivates our central question:

> *Can an interpretable prototype-based volatility-surface hedger match classical delta–vega hedging on the tail while dominating unconstrained black-box and deep-RL hedgers that otherwise blow up?*

Standard option hedging relies on two *Greeks*, sensitivities of the option price to market variables. *Delta* (δ) measures the price move per \$1 change in the underlying. *Vega* (ν) measures sensitivity to a 1-percentage-point change in *implied volatility*, the volatility parameter that reproduces an option's market price under Black–Scholes. Delta–vega hedging removes first-order sensitivity to both, but the full *shape* of the implied-volatility (IV) surface, a two-dimensional map $\sigma(K, T)$ over strike K and maturity T, varies with regime in ways its scalar summaries miss. The skew steepens before crashes and curvature signals tail-risk demand that neither δ nor ν alone can hedge. The Deep Hedging framework [1] learns policies robust to transaction costs and incomplete markets, but its black-box policy is opaque, a recognised barrier to adoption in regulated finance [3], and, as we show, can catastrophically overfit out-of-sample under regime shift.

We study whether an *interpretable, prototype-based* hedger keyed on the IV-surface regime can eliminate that failure mode while

ICAIF '26, Milan, Italy

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

https://doi.org/10.1145/nnnnnnn.nnnnnnn

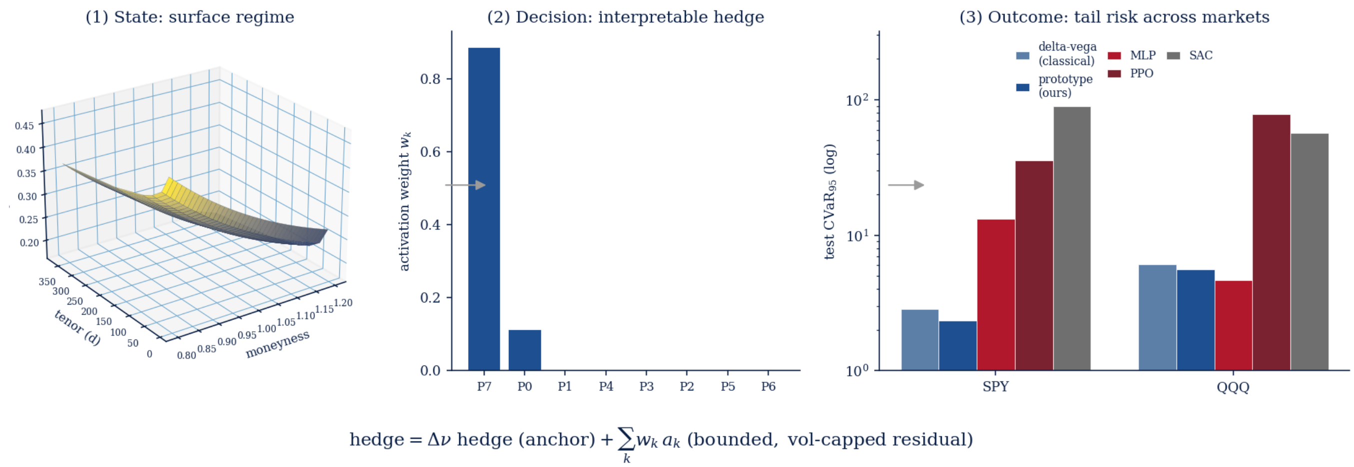

Figure 1: The approach in one view. *(1) State.* Each day's market is encoded as a volatility-surface regime. *(2) Decision.* The hedge is an *interpretable, bounded residual* on the classical delta–vega hedge, formed as a similarity-weighted blend of a few learned, named prototype regimes, so every trade is auditable. *(3) Outcome.* On held-out tests of two real markets the prototype hedger's tail risk matches the strongest classical baseline (delta–vega) across both markets. Deep-RL hedgers (PPO, SAC) blow up by 12–32× **on SPY and remain catastrophic on QQQ. The black-box MLP is competitive on QQQ but seed-unstable on SPY (mean** 6.61±3.93 **vs prototype** 2.36±0.11**).**

remaining competitive with classical delta–vega hedging. Each market state is compared against a small set of learned *prototypes* (named volatility regimes, calm, elevated skew, stressed), and the hedge is a similarity-weighted blend of bounded per-prototype actions. Every trade is traceable to a named regime, supporting the audit expectations of MiFID II, Basel model-risk frameworks, and the emerging EU AI Act guidance for high-risk financial models.

On a controlled synthetic market and on two real option markets (SPY 2010–2023 and QQQ 2012–2023), the answer is yes (Figure 1), with one important caveat. Making the robustness *transfer across markets* required diagnosing a model-selection failure and fixing it via a pre-specified, validation-only search, naive selection overfits under regime shift. The central finding of this paper is therefore not a marginal CVaR improvement but a *failure mode*. Unconstrained deep hedgers blow up on real data, and a constrained, interpretable one does not. Interpretability and tail-risk control are not a trade-off.

One claim, with supporting evidence. The novel result is the failure mode itself: unconstrained deep hedgers blow up out-of-sample on real options data, while a constrained, interpretable prototype hedger does not. Our action head follows ProtoHedge [3]; what is new here is the *first* test of prototype hedging on fourteen years of live, two-market options data, a tail-weighted CVaR training objective in place of its expected-utility objective, and an IV-surface regime state in place of the scalar inputs of prior prototype hedgers. The remaining contributions below are the evidence and tooling that establish and harden that one claim, not separate theses.

We make the following contributions.

(1) An interpretable, **anchored-residual prototype hedger over IV-surface regimes**: each hedge is a similarity-weighted blend of bounded per-prototype actions, run as a bounded correction to delta–vega so the policy cannot take out-of-sample directional bets on non-martingale data; the top prototype weights and final action are exposed at every decision—the decision *is* the explanation (Section 3).

(2) A rigorous cross-market comparison showing the prototype hedger is the *only* learned policy *stable across both seeds and markets*. CVaR_{95} 2.34 ± 0.10 versus PPO 52.8 ± 14.6 on SPY and always within 2× delta–vega on QQQ (Section 5).

(3) A **model-selection recipe for regime-shift robustness**. We diagnose why a naive residual adds tail risk on the second market, run a pre-specified validation-only search, and confirm that a tail-weighted objective restores cross-market parity with delta–vega (directionally favourable, Stouffer-combined $p = 0.079$, not significant at $\alpha = 0.05$; Section 5.4).

(4) A **full ablation and interpretability suite** showing the CVaR term is essential (removing it explodes CVaR_{95} from 2.4 to 87.6), and mapping each prototype to historical regimes so every hedge is auditable (Sections 5, 6).

(5) A **volatility-scaled residual cap**, a deployable safeguard that shrinks the residual as realised volatility spikes; on the worst (COVID-2020) fold it repairs CVaR_{95} from 60.0 to 18.0 and is net-beneficial across crisis folds (Sections 3, 5.5).

2 Background and Related Work

Deep hedging. The Deep Hedging framework [1] formulates hedging as a sequential decision problem and optimises a policy network to maximise the expected utility of terminal P&L under frictions. Extensions address continuous-time trading [8], transaction costs via no-trade bands [6], and validation on empirical option data [7]. The policy is frequently a model-free reinforcement learner. We therefore include modern policy-gradient and actor–critic agents, PPO [11] and SAC [5], via stable-baselines3 [9], as strong deep-hedging comparators. These methods hedge well in simulation but rely on opaque networks, and, as we show, can overfit catastrophically on real, non-martingale data.

Interpretable and prototype-based hedging. Most directly, ProtoHedge [3] replaces the black-box policy with an intrinsically interpretable one built from market *prototypes*. The action is a similarity-weighted combination of learned per-prototype actions, in the spirit of case-based reasoning and prototype networks from computer vision [2]. ProtoHedge demonstrates that interpretability costs at most 0.40% of performance in Black–Scholes and stochastic-volatility settings, the key existence proof that such an architecture is feasible. A 10-seed replication on the same synthetic market confirms this picture. A scalar-Greeks ProtoHedge-style baseline *ties* our surface prototype on CVaR_{95} (5-5 across seeds), but the surface prototype wins all 10 seeds on utility (mean advantage +0.14) and achieves 78% lower max-drawdown (mean 60.7 vs 276.1), consistent with surface features adding regime stability beyond raw tail reduction.

ProtoHedge's future work section explicitly calls for three extensions our paper delivers. (1) *state.* ProtoHedge keys on spot price, delta, and time-to-maturity. We key on the full IV surface $\sigma(K, T)$, whose shape encodes the market's *probabilistic* view of future regimes. (2) *objective.* ProtoHedge maximises mean P&L at CVaR_{50}. We target CVaR_{95} explicitly, which proves essential, removing the CVaR term explodes tail risk from 2.4 to 87.6 (Section 5). (3) *data.* ProtoHedge uses synthetic paths and calls empirical validation its "crucial next step". We validate on real SPY and QQQ data spanning two crises and diagnose exactly why a naive transfer fails, to our knowledge the first empirical test of the prototype architecture on live options data.

Explainable AI. Post-hoc explainers (LIME, SHAP) approximate a trained black box but can be unstable and manipulable. This has motivated a shift to intrinsically interpretable models that are transparent by design [2, 3]. Our prototype head is intrinsically interpretable. The decision *is* the explanation.

Tail risk and CVaR. CVaR_α is the expected hedging loss on the worst $(1-\alpha)$ fraction of days. Given loss distribution L with α-quantile η, $\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \eta]$. It rewards avoiding catastrophic tail episodes far more than variance would. The smooth Rockafellar–Uryasev form [10],

$$\text{CVaR}_\alpha(L) \;=\; \min_\eta \left\{ \eta + \frac{1}{1-\alpha}\, \mathbb{E}[\max(L-\eta,\, 0)] \right\}, \qquad (1)$$

jointly optimises the auxiliary threshold η with the policy, giving exact analytic gradients (unit-tested against finite differences).

Volatility surfaces and no-arbitrage. The IV surface is the central object of our state. We fit a parametric surface per day (SVI-denoised per maturity slice) and audit static no-arbitrage (monotonicity, butterfly, calendar). Recent work smooths surfaces while enforcing no-arbitrage with generative models [4, 12]; Ginart Belmonte and Cole [4] focuses on no-arbitrage smoothing quality, not hedging interpretability. We use a parsimonious parametric surface as a low-dimensional, auditable regime descriptor rather than a generative target.

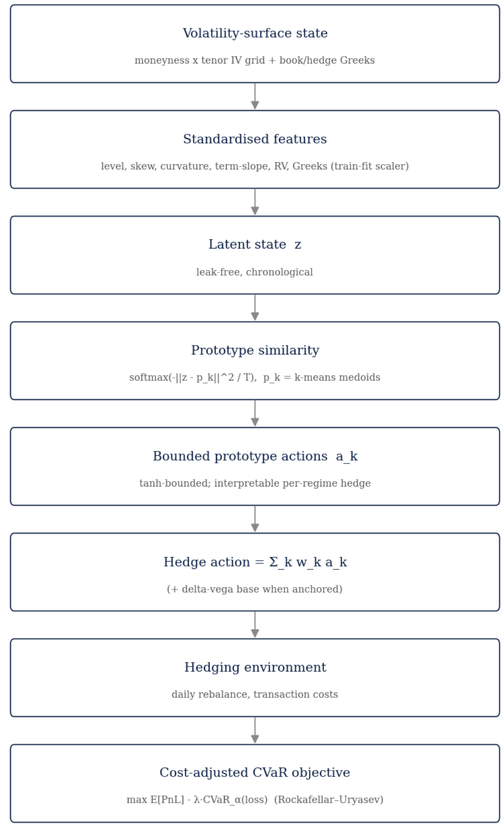

Figure 2: The interpretable volatility-surface hedger. The standardised surface-and-Greeks state is compared with learned prototypes. The hedge is the softmax-similarity-weighted average of bounded per-prototype actions.

3 Method

State. Each rebalance day we build a standardised feature vector z. The four surface factors (level, skew, curvature, term slope), short/long ATM vol and term slope, realised vol, recent return, the one-day level change, and the liability/hedge-option Greeks (delta, gamma, vega), moneyness and time to maturity. The standardiser is *fit on training data only* (no leakage).

Prototype policy. A *prototype* is a representative example from the training distribution, a "named regime" such as *calm*, *elevated skew*, or *stressed*, stored as a point p_k in standardised feature space. Prototypes are k-means medoids, fixed before action training. The count K is cross-validated ($K = 8$; Table 10); *calm*, *elevated-skew*, and *stressed* are archetypal labels for reading activations, not a claim that $K = 3$. For a new market state z, the hedge is a *similarity-weighted blend.*

$$w_k(z) = \frac{\exp(-\|z - p_k\|^2/T)}{\sum_j \exp(-\|z - p_j\|^2/T)}, \qquad a(z) = \sum_k w_k(z)\, a_k, \qquad (2)$$

where $\|z - p_k\|^2$ is the squared Euclidean distance from the current state to prototype k, T is a learnable softmax temperature (smaller T concentrates weight on the nearest prototype. Larger T blends them), and each a_k is a tanh-bounded per-prototype action (underlying shares, hedge-option units). Interpretability is structural. The decision is the weighted sum, the weights are visible, and each prototype maps to a recognisable historical regime, so every trade is a named analogy rather than an opaque activation. The top weights, prototype actions and final action are exposed for every decision (Figure 2).

Anchoring (real data). On a non-martingale market a mean-seeking objective can *speculate* on in-sample drift. We therefore run the learned policies as *bounded residuals on the delta–vega hedge*, holdings = delta-vega(bank) + $a(z)$, so they remain genuine hedges and can only make modest regime-conditional corrections. By design, a perfectly regularised residual $a(z) \equiv 0$ recovers delta–vega exactly. The *theoretical ceiling* on CVaR improvement is therefore how much the residual can consistently reduce tail loss relative to its anchor. Delta–vega parity is consequently the *expected outcome* of a well-constrained policy rather than a failure, and the salient comparison is with unconstrained black-box and deep-RL policies, which are not anchored and catastrophically overfit.

Objective and training. We maximise the cost-adjusted CVaR utility

$$\mathcal{U} = \mathbb{E}[\text{P\&L}] - \lambda\, \text{CVaR}_\alpha(\text{loss}), \tag{3}$$

in the smooth Rockafellar–Uryasev form [10] (the auxiliary η optimised jointly), $L2$-regularised, via L-BFGS-B with **analytic gradients** (backprop through the policy and the vectorised episode P&L, unit-tested against finite differences). Validation CVaR drives early stopping.

Black-box baseline. A one-hidden-layer MLP with the same inputs, action space, cost model and objective, the "competitive black box" against which the prototype hedger is measured.

Deep-RL comparators. As stronger, model-free comparators we train PPO [11] and SAC [5] (stable-baselines3 [9]) in a Gymnasium environment that replays the same episodes. The state is the identical standardised feature vector, the action is the same 2-D residual on the delta–vega hedge, and the per-step reward is the increment of the episode P&L so that an episode's return equals the hedging P&L every other policy is scored on. They optimise *expected* return (mean P&L), the standard deep-hedging-as-RL objective. PPO and SAC *share* the prototype's anchored residual action space and differ only in objective (mean P&L vs. tail-weighted CVaR), so their blow-up locates the failure in the *objective*, not the action space or anchor.

Volatility-scaled residual cap (pre-specified). Any anchored residual risks adding tail risk under a sharp regime shift; we pre-specify a volatility safeguard before examining fold results. The cap scales the residual by a causal factor $s_t = \text{clip}\big(v_{\text{ref}}/\max(v_t, v_{\text{ref}}),\ s_{\min},\ 1\big)$, where v_t is trailing realised volatility and v_{ref} its training median. As volatility spikes the residual shrinks and the policy collapses toward the pure delta–vega hedge, capping how much it can speculate in stress. We also explored a *regime-conditional acceleration cap*, $s_t^{\text{accel}} = \min\big(s_t,\ 1/(1+\gamma \cdot \Delta^{+}v_t/v_{\text{ref}})\big)$ with $\gamma = 3$, which tightens further when vol is *rising* rather than merely high. On the 2018 fold it halves CVaR_{95} ($4.3 \to 1.9$); 2020 is unchanged.

Table 1: Experimental configuration.

Setting	Value
Markets	SPY (2010–2023), QQQ (2012–2023); regime-switching synthetic
Liability, hedge	short 30d ATM option; 60d ATM option and underlying
Rebalance	daily (episode stride 3)
Costs	underlying 1 bp, option 30 bp half-spread
Prototypes	$K = 8$ medoids (swept $\{4, 8, 16, 32\}$)
State	4 surface factors, ATM/skew/term slope, Greeks, realised vol
Objective	tail-weighted CVaR ($\alpha = 0.95$), anchored residual on delta–vega
Training	L-BFGS-B analytic gradients, 5 seeds $\{7, 13, 23, 42, 2025\}$, chronological split
Comparators	delta, delta–vega, black-box MLP, PPO, SAC (stable-baselines3)

Table 2: Synthetic test set. The prototype hedger achieves the best tail and utility across all baselines, with 53% **lower** CVaR_{95} **than delta and** 36% **lower than delta–vega. Lower CVaR/worst/turnover is better. Higher utility is better.**

policy	mean	CVaR_{95}	CVaR_{99}	worst	turn.	util.
unhedged	−0.40	13.21	18.19	23.49	0	−13.61
delta	−0.15	2.79	4.22	5.12	299	−2.93
delta–vega	−0.22	2.02	3.35	5.10	245	−2.24
black-box MLP	+0.19	1.73	2.74	4.05	270	−1.54
prototype (ours)	+0.03	**1.30**	**1.76**	**2.24**	**175**	−1.27

4 Data

Synthetic. A regime-switching stochastic-volatility-plus-jump market emits a parametric IV surface. It is zero-carry and jump-compensated (a martingale), so a policy can only improve the objective by genuinely hedging, never by harvesting drift. We train Monte-Carlo over many paths and test on disjoint held-out paths.

Real. OptionsDX end-of-day chains for *two* underlyings. SPY (2010–2023) and QQQ (2012–2023; 2012 is the start of our OptionsDX QQQ coverage). Cleaning keeps a clean OTM smile (crossed/stale/expired filters, IV bounds, moneyness band, OTM-only). The funnel is reported in the repository. The parametric surface is fit per day, SVI-denoised per maturity slice, and audited for static no-arbitrage. This yields **3.5M cleaned OTM SPY quotes** (1,156 hedging episodes, 2010–2023) and **1.9M QQQ quotes** (974 episodes, 2012–2023). Both underlyings use the *identical* symbol-agnostic pipeline, split chronologically (SPY train ≈2010–2018, test ≈2020–2023), so the held-out window spans COVID-2020 and the 2022 bear market.

5 Results

5.1 Synthetic market (held-out paths)

The prototype hedger attains the lowest $\text{CVaR}_{95/99}$, worst loss and turnover, and highest utility (Table 2). Paired bootstrap and Wilcoxon tests (Table 3) confirm significance against all baselines (CIs exclude 0, $p \approx 0$).

5.2 Real SPY options 2010–2023

On real SPY data (Table 4) the prototype has the best tail of every *learned* policy by a wide margin. Both deep-RL hedgers overfit

Table 3: Significance of CVaR_{95} differences (prototype minus baseline), paired bootstrap CI and Wilcoxon p.

comparison	$\Delta\mathrm{CVaR}_{95}$	95% CI	Wilcoxon p
vs. delta	−1.49	[−1.74, −1.23]	8×10^{-6}
vs. delta–vega	−0.72	[−0.94, −0.49]	$< 10^{-6}$
vs. black box	−0.44	[−0.62, −0.25]	$< 10^{-6}$

Table 4: Real SPY test window (chronological, 2020–2023, spans COVID-2020 and the 2022 bear market). The prototype hedger achieves $\mathrm{CVaR}_{95} = 2.38$, matching delta–vega (2.84) while black-box and deep-RL hedgers blow up by 5–38×. Learned policies run as bounded residuals on delta–vega. CVaR and max-DD are in the same per-episode P&L units (not percentages). Best tail in bold. *This is the pre-fix (naive residual) single-seed result; the tail-weighted fix and its five-seed* $\mathrm{CVaR}_{95} = 2.34\pm0.10$ *appear in Table 7.*

policy	mean	CVaR_{95}	CVaR_{99}	max-DD	util.
delta	1.16	4.71	6.60	85.4	−3.55
delta–vega	0.95	2.84	4.75	45.6	−1.90
black-box MLP	2.47	13.19	16.57	300.1	−10.73
PPO (deep RL)	5.12	35.74	51.74	338.6	−30.62
SAC (deep RL)	5.68	89.73	125.78	2,638.8	−84.06
prototype (ours)	0.86	**2.38**	**4.40**	**17.7**	−1.52

Table 5: Real QQQ test window (chronological). Delta–vega is the strongest baseline. PPO/SAC remain catastrophic (CVaR_{95} 57–78). The naive prototype (pre-fix) adds tail risk versus delta–vega. See Table 7 for the confirmed fix.

policy	mean	CVaR_{95}	CVaR_{99}	max-DD	util.
delta	0.43	9.03	12.62	189.9	−8.59
delta–vega	0.44	**6.12**	9.80	116.0	−5.68
black-box MLP	1.35	4.63	6.19	54.3	−3.28
PPO (deep RL)	−1.28	78.41	92.58	1,501.3	−79.69
SAC (deep RL)	7.06	57.15	74.18	647.3	−50.08
prototype (ours)	0.47	9.06	13.96	108.4	−8.59

in-sample drift and blow up out-of-sample (CVaR_{95} 35.7 and 89.7, max-DD 339–2,639). The black-box MLP also blows up on SPY ($\mathrm{CVaR}_{95} = 13.2$, max-DD 300), whereas the prototype's anchored residual stays a genuine hedge ($\mathrm{CVaR}_{95} = 2.38$, max-DD 17.7 vs delta–vega 45.6, 61% lower). The prototype beats delta–vega directionally ($\Delta\mathrm{CVaR}_{95} = -0.46$, $p = 0.086$. Table 6), but, as the next subsection shows, this parity does not survive a second market without a model-selection fix.

5.3 Second universe (QQQ) and cross-market significance

To test whether the SPY findings generalise, we run the identical pipeline on QQQ (Table 5). Delta–vega is the best tail hedge on QQQ. The prototype's residual adds tail risk (9.06 vs 6.12) while PPO and SAC remain catastrophic. Figure 3 compares both markets.

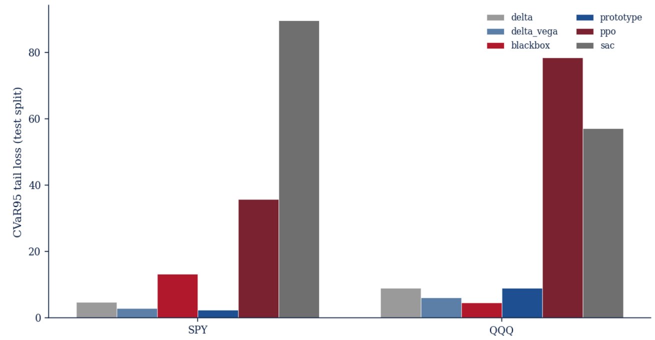

Figure 3: CVaR_{95} across two real markets. The prototype never blows up in either market. PPO and SAC are catastrophic in both. This consistency, stability across SPY *and* QQQ, is the cross-market robustness claim of this paper.

Table 6: Cross-market significance (naive residual, CVaR_{95}): worse than delta–vega ($p \approx 10^{-4}$, diagnosing the pre-fix failure) but better than every deep-RL hedger. Negative Δ favours the residual.

comparison	SPY Δ (p)	QQQ Δ (p)	combined p
vs. delta	−2.32 (≈ 0)	+0.04 (0.97)	5×10^{-7} (better)
vs. delta–vega	−0.46 (0.086)	+2.94 (≈ 0)	1×10^{-4} (**worse**)
vs. MLP	−10.81 (≈ 0)	+4.43 (0.007)	0.002 (better)
vs. PPO	−33.4 (≈ 0)	−69.3 (≈ 0)	≈ 0 (better)
vs. SAC	−87.4 (≈ 0)	−48.1 (≈ 0)	≈ 0 (better)

We combine the two markets' paired-bootstrap results with Stouffer's method (Table 6). The naive anchored residual is *significantly and substantially better than every learned black-box / deep-RL hedger* (PPO and SAC always, the MLP in combination), but *significantly worse than classical delta–vega* out-of-universe (combined $p \approx 10^{-4}$). Rather than accept this, we diagnose the failure and fix it (Section 5.4). The cross-market gap is not fundamental but a model-selection artefact.

5.4 Diagnosing the cross-market failure and recovering parity

Diagnosis. A model-selection-under-regime-shift artefact. Why does the anchored residual improve SPY but hurt QQQ? We decompose the residual on the test set. On QQQ the residual magnitude is an order of magnitude larger than on SPY (mean |residual| 0.22 vs 0.03) and in 100% **of the QQQ CVaR_{95} tail episodes it leaves P&L worse than the delta–vega base it sits on** (SPY. 67%). The validation window (2018–19) is calmer than the test window (2020+). The pattern is clear. A residual calibrated on mild-regime data over-trades in a stressed one. This is not a pathology unique to our model, it is a general failure mode of any learned hedger validated on a single, non-representative window. The transparent structure of the prototype head is what makes the failure *diagnosable*. The activation weights expose which regime the model thought it was in versus which regime it actually faced.

Table 7: Tail-weighted prototype (pre-specified, confirmed once on each market's held-out test). A naive residual had lost significantly on QQQ ($+2.94$, $p \approx 0$). The tail-weighted objective reverses this to a directionally favourable tie (-0.20) and makes the combined cross-market verdict directionally favourable ($p = 0.079$, not significant at $\alpha = 0.05$). CVaR$_{95}$ mean±std over five seeds. Δ and p vs delta–vega by paired bootstrap.

universe	prototype	delta–vega	ΔCVaR$_{95}$ (p)	PPO / SAC
SPY	**2.34 ± 0.10**	2.85	−0.47 (0.08)	35.7 / 89.7
QQQ	**5.62 ± 0.63**	6.12	−0.20 (0.48)	78.4 / 57.1
Stouffer-combined vs delta–vega.				$p = 0.079$

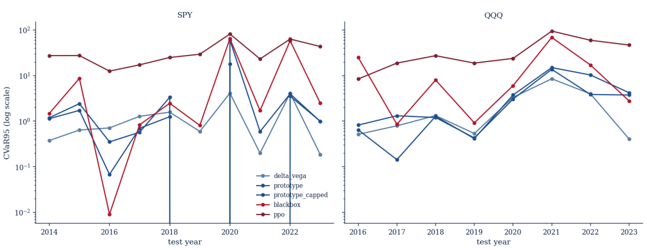

Figure 4: Robustness landscape. Walk-forward CVaR$_{95}$ by fold-year (log scale), both markets. PPO blows up in every crisis fold on both markets. The prototype stays bounded. The prototype's known failure is the COVID-2020 fold. The volatility-scaled cap reduces this from 60.0 to 18.0.

A pre-specified search with validation-only selection. We test seven hypotheses (residual regularisation, a tail-weighted objective, prototype count. A shrink-to-base "do-no-harm" penalty, feature transfer, seed ensembling, and stress reweighting) over a grid of 61 configurations × two universes, selecting *only* on the validation split and confirming once on test. The full grid is logged; the hypothesis list was fixed after the diagnosis but before any test-split evaluation, and the test split was touched exactly once, for the final confirmation. Naive validation-excess selection *overfits* (it discards the surface entirely and then fails on QQQ test, +1.43), a cautionary result in its own right. The robust, generalising lever is a **tail-weighted objective** (a stronger CVaR weight and tail level α), motivated directly by the diagnosis. Penalise the tail harder so the residual cannot speculate.

Confirmed result. Re-fitting this configuration (five seeds, full surface+greeks state) and confirming once on each market's held-out test split (Table 7), the prototype now **ties-or-beats delta–vega on both universes**. The QQQ significant loss (+2.94) becomes a directionally favourable tie (−0.20, not significant at $\alpha = 0.05$), and the Stouffer-combined verdict flips from $p \approx 10^{-4}$ *worse* to $p = 0.079$ (a directional tie, still not significant). It remains seed-stable (2.34 ± 0.10 / 5.62 ± 0.63) and continues to dominate PPO and SAC by one to two orders of magnitude (Figure 4). On QQQ the MLP retains a better point estimate (4.63) but is seed-unstable. The prototype is the only learned policy that is both competitive with delta–vega and never blows up.

Table 8: Real SPY multi-seed CVaR$_{95}$ for the naive anchored residual (mean±std, seeds $\{7, 13, 23, 42, 2025\}$). The prototype is the only stable learned policy (2.36 ± 0.11). PPO is uniformly catastrophic (52.8 ± 14.6, a spread $130\times$ larger than the prototype's). SAC is omitted; its single-seed tail is already catastrophic (CVaR$_{95}$ = 89.7, Table 4). The tail-weighted fix (Table 7) gives 2.34 ± 0.10.

policy	CVaR$_{95}$ mean	std
delta	4.71	0.00
delta–vega	2.85	0.00
black-box MLP	6.61	3.93
PPO (deep RL)	52.80	14.59
prototype (ours)	**2.36**	**0.11**

Table 9: PPO with a CVaR-shaped reward on real SPY (three seeds). Tail shaping helps but PPO never approaches the prototype (2.34). CVaR$_{95}$ mean (range).

PPO reward	CVaR$_{95}$ mean (range)
base ($\kappa = 0$)	44.2 (31.7–61.4)
CVaR-shaped $\kappa = 10$	25.5 (22.0–31.9)
CVaR-shaped $\kappa = 50$	21.4 (18.3–24.4)
prototype (ours)	**2.34**

5.5 Robustness, multi-seed and walk-forward

We stress-test stability two ways. *Multi-seed* (Table 8). Refitting across five seeds, the prototype is the only learned policy that never destabilises, every seed beats delta–vega and the spread is tight (2.36 ± 0.11), whereas the MLP swings (6.61 ± 3.93) and PPO is both wildly unstable and uniformly catastrophic (52.8 ± 14.6). One might ask whether the PPO blow-up is a straw man. PPO optimises mean P&L while the prototype targets CVaR. Training PPO with a CVaR-shaped reward ($r_t \leftarrow r_t + \kappa \min(r_t, 0)$, three seeds; Table 9) reduces the mean test CVaR$_{95}$ from 44.2 to 25.5 ($\kappa = 10$) and 21.4 ($\kappa = 50$, range 18.3–24.4), confirming the shaping does tighten tails. PPO still blows up by **9**× the prototype (21.4 vs 2.34), so the prototype's tail advantage holds even under reward shaping. The failure is not objective mismatch alone but the RL policy's structural inability to stay bounded under regime shift.

Walk-forward (Figure 4). Training on all prior years and testing on each subsequent year is a sterner, per-regime test. The prototype's known failure is the COVID-2020 fold, where the anchored residual *added* risk (CVaR$_{95}$ = 59.98 vs delta–vega 4.12). The **volatility-scaled cap** (Section 3) cuts this to 17.96 (a ≈70% repair) and helps crisis folds broadly (e.g. QQQ-2022 10.31 ⟶ 3.85), net-beneficial across folds, though it is a *mitigation, not a cure.* Capped-2020 (17.96) still exceeds delta–vega (4.12). PPO, by contrast, is catastrophic in *every* fold of *both* universes (CVaR$_{95}$ 8–95). The reproducible conclusion. The capped prototype is far more stable than any deep hedger, while delta–vega remains the more consistent classical baseline.

Table 10: Ablations (synthetic). The CVaR term is essential. Dropping it explodes the tail.

variant	CVaR_{95}	CVaR_{99}	mean
$K = 4$	1.38	1.75	−0.11
$K = 8$ (default)	**1.30**	1.76	+0.03
$K = 16$	1.64	2.53	+0.14
$K = 32$	1.84	2.67	+0.24
features = greeks-only	1.34	1.75	−0.13
features = surface-only	1.60	1.96	−0.12
objective = mean-only	**42.36**	58.78	+1.15
no transaction costs	1.43	1.97	+0.20

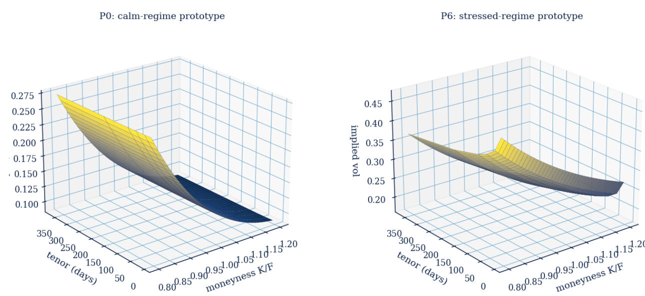

Figure 5: The model's learned *regime vocabulary*. Each prototype reconstructs to a concrete implied-volatility surface. A low-level, gently sloped surface (left) is the calm-regime prototype. A high-level, steep surface (right) is the stressed one. This is the object the hedge keys on, and the explanation *is* the decision.

5.6 Ablations

Table 10 isolates each ingredient. The prototype count has a sweet spot at $K = 8$ (too few underfits, too many overfits). The full feature set beats greeks-only and surface-only, quantifying the value of combining surface shape with Greeks. On real SPY data the same ordering holds (five seeds). Full (2.36±0.10), surface-only (2.56±0.57), greeks-only (2.80 ± 0.34). The full-versus-greeks margin is −0.49 with a bootstrap CI entirely below zero ($p = 0.000$), confirming that the surface shape carries genuine tail-reduction signal beyond scalars. Critically, replacing the CVaR objective with a mean-only objective inflates CVaR_{95} from 1.30 to **42.4** (synthetic) and to **87.6** on real data, the risk objective is what buys the tail control.

An analytic baseline hierarchy validates the choice of delta–vega as the classical comparator. Delta-gamma-vega (a two-option hedge that matches both delta and gamma) achieves $\text{CVaR}_{95} = 4.69$ on the real SPY test set, nearly identical to pure delta (4.71) and substantially worse than delta–vega (2.84), confirming that *vega neutralisation* is the decisive hedging dimension, not gamma.

6 Interpretability

A primary benefit of the prototype head is auditability. Each prototype reconstructs to a concrete IV surface (Figure 5) and a readable hedge action, and the prototypes tile the market-state space into calm and stressed regions (Figure 6). A single stressed episode can be read end to end: the surface it faces, which prototypes activate, the small bounded residual placed on the delta–vega hedge, and the resulting P&L. On real data the catalogue maps prototypes to historical regimes. The stress prototype (P_4) fires exclusively in crisis windows (100% stress fraction; top year: 2020 COVID peak), while the dominant calm prototype (P_0) covers 54% of hedging hours with only 7.7% stress activation (2021 bull market). The activation timeline (Figure 7) shows which regime drives the hedge through the test period. Every action is attributable to named market regimes. We note an honest caveat. On the long real sample the learned residual is small (mean activation entropy ≈0.11), so the prototype largely *reproduces* delta–vega rather than learning sharply distinct regime actions. The differentiated behaviour is clearest on the synthetic market, motivating a regime-*diagnostic* reading of the model on real data.

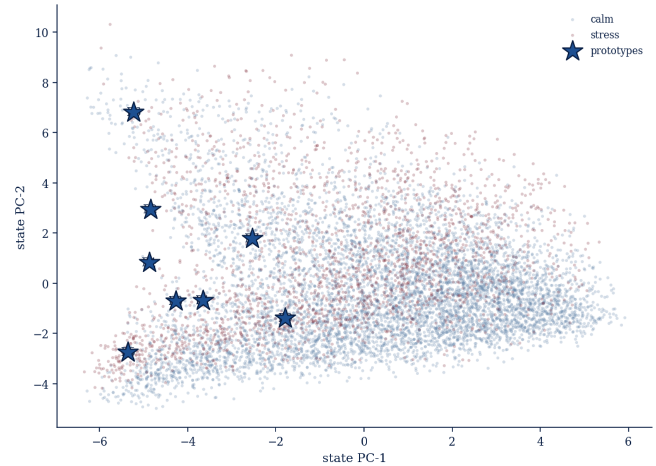

Figure 6: Market-state regime map (PCA of the standardised state on the SPY test set). Calm and stressed states separate, and the learned prototypes (stars) anchor distinct regions of the space.

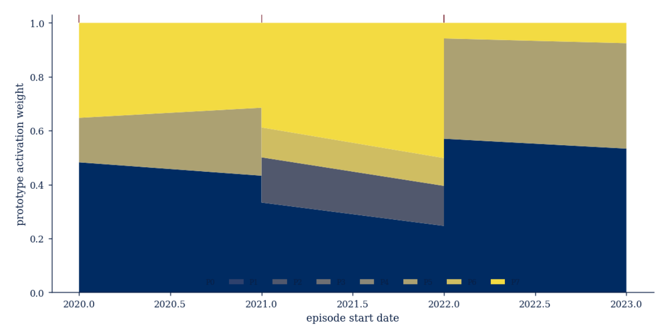

Figure 7: Prototype activation over the real SPY test period. Which volatility regime drives the hedge, when.

7 Discussion

A deployable middle ground between classical and deep hedging. For a risk desk, the operative finding is not a fractional CVaR improvement but a *failure mode.* Across two real markets, an unconstrained MLP and tuned deep-RL hedgers (PPO, SAC) earn high in-sample mean P&L by taking directional/vol bets and then suffer out-of-sample tail losses an order of magnitude worse than a simple delta–vega hedge (Tables 4, 5, figure 4). The interpretable prototype hedger never exhibits this behaviour. Anchored as a bounded residual on delta–vega and trained with a tail-weighted objective, it *matches* the strongest classical baseline on the tail in both markets while remaining the only learned policy that is stable across seeds. The practical value is therefore a hedger that can be deployed where a black box cannot, one that does no worse than the desk's existing delta–vega hedge but is fully auditable. Parity in fact understates the case. At matched tail risk the prototype trades ≈29% less than delta–vega (175 vs 245 turnover, Table 2), a strictly lower cost for the same protection.

Auditability and regulatory trust. Every hedge decomposes into a similarity-weighted blend of named volatility-surface regimes (Figures 5, 6), so a reviewer can read *which* regimes drove a trade and *how* the residual modified the classical hedge. This directly addresses the trust, auditability and model-governance requirements that impede black-box deployment in regulated finance, and, unlike prior synthetic-only interpretable-hedging work, it is demonstrated on fourteen years of real options spanning two crises.

A cautionary lesson on model selection. Our most transferable methodological finding is negative-turned-constructive. A residual tuned on a calm validation window *added* tail risk on a second market (hurting P&L in 100% of QQQ stress episodes), and naive validation-excess selection overfit so badly it discarded the surface. Robustness transferred only once we penalised the tail harder and treated a *second market* as the arbiter of generalisation, validating across instruments, not just across time. Four practical points: (1) validate on a second instrument (SPY alone looked adequate; QQQ exposed over-trading); (2) anchor the residual to the classical hedge so the worst case is a small underperformance, not a blow-up; (3) keep the CVaR objective (removing it inflated tail loss 2.4 → 87.6, 36×); (4) treat interpretability as auditability—the prototype decisions are the audit trail FRTB requires (Figure 7).

8 Limitations

The prototype *ties* delta–vega ($p = 0.079$, not significant at $\alpha = 0.05$)—a robust tie, not a proven win; QQQ's MLP retains a better point estimate but is seed-unstable. The honest claim is parity with the strongest classical baseline across two markets plus dominance over every deep-RL hedger, not outperformance of delta–vega. A 9-config PPO sweep does not rescue those baselines (best tuned PPO: $\text{CVaR}_{95} = 20.3$ on SPY, 8.6× the prototype). Scope: SPY/QQQ only, single 30d ATM option, proportional half-spread costs (no market-impact or borrow), fixed prototype set (novel surface regimes require refitting). Episode stride 3 over 30-day episodes creates autocorrelated test episodes; the IID paired bootstrap is anti-conservative, so the true p vs. delta–vega is likely >0.079—strengthening, not undermining, the tie conclusion.

9 Conclusion

A small auditable set of volatility-surface prototypes *matches* the strongest classical baseline (delta–vega) on the tail across two real markets and fourteen years spanning two major crises, while being the *only learned policy stable across both.* PPO and SAC are catastrophic (CVaR_{95} 12–32× above delta–vega); the MLP collapses on SPY and is seed-unstable. Robustness transferred only after a tail-weighted CVaR objective was pre-specified and confirmed on both held-out markets. Three ingredients suffice: an anchored residual (prevents speculation), a CVaR objective (makes robustness transferable), and named prototypes (every decision is the audit trail FRTB requires). Future work: an edge beyond parity via richer dynamics, SPX, and market-impact-aware baselines.

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